Day: Section :- B Date:

Q:2(i)(or)

Q:2(i)(or)

* **Green chemistry :-**

Green chemistry is the model of chemical products and processes that reduce the use of hazardous substance.

♥ **Importance :-**

* Green chemistry is helpful in understanding and reducing pollution.

* It aims to design products and processes that minimize the generation of hazardous waste.

* It ~~prot~~ promotes the uses of safer chemicals and renewable resources, reducing the environmental footprint of chemical processes.

Q:2(ii)(or)

Q:2(ii)(or)

Geochemistry	Astrochemistry
Definition	
Geochemistry is the	Astrochemistry is

Day: Date:

branch of chemistry that deals with the study of chemical composition, distribution and transformation of elements and compounds in the Earth crust.	the branch of chemistry that deals with the study of chemical processes and reactions that occur in astronomical ~~event~~ environments

Examples:-

* Rocks	* Stars
* Soils	* Planets
* Minerals	* ~~Commed~~ Commets.
* Water	* Interstellar space.

Q:2 (iii)

Science	Technology

.: Definition:-

Science is the systematic study to explore the natural world. Science intends to recognize the fundamental principles and processes of the	The integration of scientific knowledge for human needs is known as technology. Technology provides a pathway to the developement of system, techniques and tools.

natural world.

Purpose

* To explore and understand the natural phenomena.	* To apply knowledge for human needs.

Examples:-

* Scientists study the principles of photovoltaic cells to understand how sunlight can be converted into ~~ene~~ electricity.	* Technologists develope solar panels based on the scientific principles discovered.

Q:2 (iv) (or)

Q:2 (iv) (or)

* **Organic Chemistry:-** To treat diseases, organic chemists synthesize new medicines that interacts with specific targets (proteins or enzymes).

* **Physical chemistry:-** Physical chemistry is a part of ~~over~~ our everyday life. Batteries in our vehicles are buit on the principles of electrochemistry.

Day: Date:

Q:2 (v)
A :2 (v)

* Matter.
Anything that has mass and occupies space is known as matter.

States of matter:-

Solid:-
A solid has a definite shape and volume. Particles are tightly packed together in a fixed arrangement.
Such as:- Ice cube, Lego block, brick, table, Door.

Liquid:-
A Liquid has a definite volume but takes shape in it's container. Particles are close ~~to~~ together but can move past one another.
Such as:- Water, Honey, Oils, Fizzy drinks.

Gas:-
A gas has neither definite nor volume. Particles are far apart and move freely.
Such as:- Oxygen (O), Hydrogen (H), Carbondioxide (CO_2), Nitrogen (N).

Day: Date:

★ **Plasma :-**

A highly ionized gas where electrons are seperated from atoms. These are often found in high-energy environment. Plasma are electrically conductive.

Such as :- The lightining, the solar, wind, welding arc.

— Q:2(vi)(or) —

— Q:2 (vi)(or) —

★ **Plasma :-**

A highly ionized gas where electrons are separated from atoms. These are often found in high-energy environment. Plasma are electrically conductive.

Such as:- The welding arc, Lightining, solar, wind etc.

— Q:2(vii) —

— Q:2 (vii) —

Compounds	**Mixture**
Combinations	
★ Elements are chemically combined in a fixed ratio	Substances are physically mixed, not in a fixed ratio.

Day: Date:

Separation

* It can only be separated into through chemical reactions.	It can be separated by physical means (filtration, evaporations etc.

Properties

* It has different properties from it's constituents elements.	It's components retain their individual properties.

Q:2(viii)

Q:2(viii)

* **Allotropy:-**

The property of an element to exist in different physical form is called allotropy. These different forms in the same physical state are called allotropes. Atoms of the same element are arranged in different manners in allotrope.

Three allotropes of carbon:-

* **Diamond:-**

The hardest and purest crystalline allotrope of carbon. ~~Eat~~ Each C-atom is bonded to four other

Day: Date:

carbon atoms forming a rigid tetrahedral network. Non-conductor of electricity.

✓ **Graphite:-**

It is composed of flat two-dimensional layers of hexagonally arranged carbon atoms. Weak intermolecular bonds between layer make it soft and slipery. It is a good conductor of electricity.

✓ **Buckyballs (C-60):-**

It is also known as fullerenes. It have a football-fused hollow ring structure made up of twenty hexagons and 12 pentagons. Each of it's 60 carbon atoms in bonded to 3 other carbon atoms.

—— Q:2(ix) ——

—— Q:2(ix) ——

* **Solution:-**

A solution is a homogenous mixture of ~~two~~ two or more substances in which one ~~substance~~ a substance is dissolved in the other. Homogenous ~~solution~~ means no particles or parts of different substances

Day: Date:

can be seen. The particles in solution are microscopic.

FOR EXAMPLE:-

Salt is dissolved in water.

* **Solute :-**

A ~~solute~~ substance that is dissolved is called solute.

FOR EXAMPLE:-

Salt.

* **Solvent :-**

A substance ~~that is~~ in which the solute is dissolved is called solvent.

FOR EXAMPLES:-

Water.

Q:2(x)

Q:2(x)

Saturated solutions	Unsaturated solution.

Definition

A solution containing maximum amount of solute at a given temperature.	A solution containing lesser amount of solute than that which is required to saturate it at a given temperature.

Saturated	Unsaturated
Dissolving	
* It cannot dissolved any more ~~sa~~ solute; If further added solute remains undissolving	It has capacity to dissolve more solute.
Equilibrium	
* Dynamic Equilibrium is established between dissolution and crystallization	No equilibrium; more solute can be dissolved.

Q:2(xi)

Q:2(xi)

Colloid	Suspension
Particle size	
* 1 to 1000 nm. It is not visible by naked eye.	It is greater than 1000 nm (10^3 nm). Visible by naked eye.
Filtration	
* Particles can pass through normal filter paper but not through ultra filter paper.	Particles can passnot through normal filtra paper as well as ultra filter paper.

Day: Date:

Stability

It does not separate on standing	It can only separates or settle down when stationary.

Examples:-

* Milk	* Chalk in water
* Starch	* muddy water
* Blood	* paints.

Section - C

Q: 3(i)

Q: 3(i)

✓ **Organic Chemistry:-**

Organic chemistry is the branch of ~~organ~~ chemistry ~~that~~ that deals with ~~the study of~~ substances containing organic compounds (except carbonates, bicarbonates, carbides and oxides)

Daily life application:-

To treat ~~organic~~ diseases, organic chemists synthesize new medicines that interact with specific targets like (proteins or enzymes).

Day: Date:

Inorganic Chemistry :- Inorganic chemistry is the study of chemistry that deals with elements and their compounds except organic compounds.

Daily Life ~~of~~ application:- ~~Lit Lith~~

Lithium-ion (Li-ion) batteries are used ~~&~~ as rechargeable batteries for electronics, toys, wireless headphones, handheld power tools, small and large appliances, electrical storage ~~derics~~ devices and electric vehicles.

Physical chemistry :- Physical chemistry is the branch of chemistry that deals with ~~the~~ ~~study of~~ the laws and ~~theory~~ theories to understand the structure and changes of matter.

Daily life application : Physical chemistry is a part of our everyday life. The batteries in our vehicles that are built in the principles of electrochemistry.

Day: Date:

Analytical chemistry:-

Analytical chemistry is the branch of chemistry that deals with the methods and instruments ~~that~~ for determining ~~the~~ the composition and properties the matter.

Daily life application:-

Forensic chemistry is the application of analytical chemistry. It involes the examination of physical traces, such body fluids, bones, fibers and drugs. It can be used to identify an unknown compound. For Example drugs are often found in various ~~com~~ coloured powders and are analyzed to determine their contents.

~~Physical~~ Environmental chemistry:-

Environmental chemistry is the branch of chemistry that deals with the study of chemical ~~processes~~ and toxic substances that occurs in ~~env~~ environment and their adverse effect on human beings.

Daily life application:-

Environmental chemistry is used to protect water, that has been ~~poll~~ polluted by soil

Day: Date:

, dust by using different methods e.g sendimentation, filtration, evaporation

Q:~~143~~(ii)

Q:4(ii)

Matter:- Anything ~~That~~ That has mass and occupies space is known as matter.

Four ~~t~~ states of matter:-

✓ **Solid:-** A solid has a definite shape and volume. Particles are tighly packed together in a fixed arrangement. Solids have high density at normal condition and they cannot flow.

Example:-

Ice

Metal

Diamonds

Graphite.

✓ **Liquid:-** A liquid has a definite volume but ~~shap~~ takes shape in every container. Particles ~~are~~ are ~~c~~ close together. but move past one another. Liquids.

Day: ______ Date: ______

high have high density and moderately compressible and can flow.

FOR EXAMPLE:-

* Fizzy drink,
* Water
* Milk.

✓ **Gas :-**

A gas have neither a definite shape nor volume. Particles are far apart and can flow very easily. Gases have low condition at normal conditions due to empty spaces between molecules

FOR EXAMPLE:-

* Oxygen (O_2)
* Hydrogen (H_2)

✓ **Plasma :-**

A highly ionized gas where electrons are separated from atoms, often found in high-energy environment Plasmas are electrically conductive.

FOR EXAMPLE:

* Lighting
* Wind
* Solar
* Sun

Day: Date:

Gas	Liquid	Solid
	Density	
Low	High	High
	Compressibility	
Very compressible	Moderately compressible	Not compressible
	Fluidity	
Can flow	Can flow	Cannot flow.

Q: 3(iii)

Q. 3(iii)

Elements:-

An element is a pure substance made up only one type of atom. It cannot be broken down into simpler substances by chemical means. Elements contain only one type of atom.

Example:-

Oxygen (O_2)
Hydrogen (H_2)
Carbon (C)

Compounds:-

A compound is a pure substance made up of two or more elements, chemically combined in a fixed ratio. Compounds have different properties from the

Day: Date:

elements that make them and can only be separated into their elements through chemical reactions. Compounds consist of two or more elements chemically bonded in fixed proportions.

Examples:

- Water (H_2O)
- Carbondioxide (CO_2)

Mixture :-

A combination of two or more substance (elements or compounds) that are physically together but not chemically combined is called ~~mixe~~ mixture. Mixtures can be separated into their components by physical means, they do not have a fixed ratio. Mixtures consist of multiple substances that retain their individual properties and can only be separated by ~~Physical~~ physical processes.

EXAMPLES:-

- Air
- Saltwater
- Sand and iron fillings.

Day: Date:

17

Matter

- Pure substances
  - Elements
  - Compounds
- mixture
  - homogenous mixture
    - Solution
      - Saturated
      - unsaturated
      - Super-saturated
  - heterogenous mixture
    - Colloids
    - suspens-ions

~~Q: 63(iv)~~

Q: 3(iv)

★ **Solutions:**

A solution is homogenous mixture of two or more components
FOR EXAMPLES:- ~~A~~ Sea water.

★ **Colloid:-**

A colloid is a heterogenous mixture in which the solute particles are larger than those in true solutions but are not large enough to seen by naked eye. Particles are range between 1 and 1000nanometre in diameter. and remain evenly distributed. They are also known as

colloidal dispersions or false solutions. The tyndall effect distinguishes colloids from solution.
EXAMPLES:- Starch, Blood.

* **Suspensions:-**

A heterogenous mixture in which solid particles are spread throughout the liquid without dissolving. Particles are big enough to ~~see~~ be seen by naked eye (greater ~~then~~ than 1000nm).
EXAMPLES:- Chalk in water
↳(milky suspension)
: Paints.

* **Detailed comparison table:-**

Solution	Colloid	Suspension
	~~Partict~~ Type	
Homogenous mixture	Heterogenous mixture	Heterogenous mixture
	Particle size	
less then 1 nm. Not visible by naked eye	1 to 1000 (1-10^3nm) Not visible by naked eye	Greater by 10^3nm. Visible by naked eye.

Day: Date:

(19)

Filtration

* Particles can move through normal as well as ultra filter	Particles can move through normal but not ultra filter	Particles cannot move through normal ~~but~~ as well as ultra filter.

Tyndall effect

* Cannot scatter the light (due to small size)	Can scatter the light (Tyndall effect)	Can scatter the light (Tyndall effect)

Stability

* Does not separate	* does not separates	* Separates or settles down ~~stationary~~ stationary

Examples

Sea water	Milk	Muddy water.

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

CHEMISTRY — SSC-I (9th Grade)

Total Marks: 65 **Time Allowed: 2 Hours 30 Minutes** **Version: 1**

General Instructions:

1. Read all questions carefully before attempting.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. *Time: 15 minutes.*
3. **Section B** (Short Questions): Attempt **11 questions**, *choosing either the main question **OR** its alternative.* Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **4 questions**, *choosing either the main question **OR** its alternative.* Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.
6. No extra time will be given for Section A after 15 minutes.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	Chemistry deals with the composition, structure, properties, and changes of:	Energy	Space	Matter	Light	Ⓐ Ⓑ ● Ⓓ
2	The branch of chemistry that deals with carbon-containing substances (except carbonates, bicarbonates, oxides, carbides) is:	Inorganic	Physical	Organic	Analytical	Ⓐ Ⓑ ● Ⓓ
3	The branch that focuses on the study of polymers, their types, properties, and uses is:	Industrial	Polymer	Nuclear	Astro	Ⓐ ● Ⓒ Ⓓ
4	The integration of scientific knowledge for human needs is known as:	Science	Engineering	Technology	Research	Ⓐ Ⓑ ● Ⓓ
5	Forensic chemistry is the application of:	Organic	Analytical	Physical	Environmental	Ⓐ ● Ⓒ Ⓓ
6	Anything that has mass and occupies space is called:	Energy	Element	Compound	Matter	Ⓐ Ⓑ Ⓒ ●
7	A highly ionized gas where electrons are separated from atoms is called:	Solid	Liquid	BEC	Plasma	Ⓐ Ⓑ Ⓒ ●
8	Bose-Einstein Condensate (BEC) occurs at temperatures near:	100°C	0°C	-273.14°C	25°C	Ⓐ Ⓑ ● Ⓓ
9	Diamond, graphite, and buckyballs are allotropes of:	Silicon	Oxygen	Carbon	Nitrogen	Ⓐ Ⓑ ● Ⓓ
10	A solution containing maximum amount of solute at a given temperature is:	Unsaturated	Super-saturated	Dilute	Saturated	Ⓐ Ⓑ Ⓒ ●
11	The scattering of light by colloidal particles is called:	Reflection	Refraction	Tyndall effect	Dispersion	Ⓐ Ⓑ ● Ⓓ
12	Water is called a universal solvent because it:	Is colorless	Dissolves most compounds	Is everywhere	Has no taste	Ⓐ ● Ⓒ Ⓓ